Skills Summary

Writing and Interpreting Circle Equations

Use this reference while completing your worksheet or when studying.

Skill 1 — Read the center and radius off an equation

Standard form: $(x-h)^2+(y-k)^2=r^2$, center (h,k), radius r.

The form *subtracts* the center, so $x+1$ means $h=-1$. The right-hand side is r^2, so take a square root to get r. A point is on the circle exactly when its coordinates make the left side equal r^2.

Example: Give the center and radius of $(x+1)^2+(y-4)^2=64$.
Step 1. Rewrite each bracket as a subtraction to read the center, then square-root the right side for the radius.
Step 2. $x+1=x-(-1)$ and $y-4$ give center $(-1,4)$; the radius is $\sqrt{64}=\mathbf{8}$.

Try it: Find the radius of $(x-5)^2+(y+2)^2=45$, to two decimal places.

check: **6.71**

Skill 2 — Write the equation from a center and radius

Recipe: substitute h, k and r into $(x-h)^2+(y-k)^2=r^2$.

Square the radius — the right-hand side is r^2, never r. If the radius is not given, it is the distance from the center to a known point of the circle.

Example: Write the equation with center $(2,-3)$ and radius 4.
Step 1. The center and radius are both given, so substitute directly and square the radius.
Step 2. $h=2$, $k=-3$, $r^2=4^2=16$, giving $\boldsymbol{(x-2)^2+(y+3)^2=16}$.

Try it: Write the equation with center $(-5,1)$ and radius 7.

check: $\boldsymbol{(x+5)^2+(y-1)^2=49}$

Skill 3 — Complete the square to reach standard form

From $x^2+y^2+Dx+Ey+F=0$:

group the x-terms and the y-terms, move F to the right, then add $\left(\frac{D}{2}\right)^2$ and $\left(\frac{E}{2}\right)^2$ to **both** sides. Whatever you add on the left must be added on the right, or the radius comes out wrong.

Example: Write $x^2+y^2-4x+2y-4=0$ in standard form.
Step 1. The equation is in general form, so group each variable's terms and complete both squares.
Step 2. Half of -4 squared is 4 and half of 2 squared is 1: $(x^2-4x+4)+(y^2+2y+1)=4+4+1$, giving $\boldsymbol{(x-2)^2+(y+1)^2=9}$.

Try it: Write $x^2 + y^2 + 6x - 8y + 9 = 0$ in standard form.

check: $\boldsymbol{(x+3)^2 + (y-4)^2 = 16}$

Skill 4 — Use the equation to answer a question

Substitute what you know.

To find where a circle meets a horizontal line $y = c$, put c in for y and solve the remaining equation for x; a positive leftover gives two crossing points, zero gives one (tangent), and a negative leftover means the line misses the circle.

Example: Where does $(x-1)^2 + (y-2)^2 = 25$ meet the line $y = 6$?
Step 1. The line fixes y, so substitute $y = 6$ and the circle equation becomes a single equation in x.
Step 2. $(x-1)^2 + 16 = 25$, so $(x-1)^2 = 9$ and $x - 1 = \pm 3$: $\boldsymbol{x = -2}$ or $\boldsymbol{x = 4}$.

Try it: Where does $(x+2)^2 + (y-1)^2 = 100$ meet the line $y = 7$?

check: $\boldsymbol{x = -10}$ or $\boldsymbol{x = 6}$

(!) Watch out: The number on the right is r^2. A circle written $=\ 20$ has radius $\sqrt{20} \approx 4.47$, and an equation whose right side comes out negative has no graph at all.